

Review

Effects of ovarian ablation or suppression in premenopausal breast cancer: A meta-analysis of randomized controlled trials



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Abstract

Background: The effect of ovarian ablation or suppression (OAS) in premenopausal women with breast cancer is controversial. The overall survival (OS), disease-free survival (DFS) and adverse event of OAS versus no OAS were compared.

Methods: A literature review of EMBASE, Web of Science, PUBMED, and Cochrane Library was conducted. The hazard ratio (HR) and 95% confidence interval (CI) for OS and DFS, as well as risk ratio (RR) and 95% CI for adverse events were evaluated. I-squared statistic (I^2) represents heterogeneity.

Results: Twenty-nine studies with a total of 21,249 women were included. In premenopausal women aged 40 years or younger, there were significant differences in OS (HR 0.78, 95% CI: 0.66–0.94, $P = 0.008$, $I^2 = 0\%$) and DFS (HR 0.84, 95% CI: 0.73–0.97, $P = 0.02$, $I^2 = 0\%$) between OAS and no OAS. In advanced stage breast cancer, a significant difference was found in OS (HR 0.76, 95% CI: 0.60–0.96, $P = 0.02$, $I^2 = 0\%$). Patients treated with OAS had more chances to have hot flushes (RR 1.91, 95% CI: 1.62–2.26, $P < 0.01$, $I^2 = 0\%$) and vaginal dryness (RR 1.19, 95% CI: 1.08–1.31, $P = 0.0003$, $I^2 = 0\%$). No significant difference in depression (RR 1.28, 95% CI: 0.94–1.74, $P = 0.12$, $I^2 = 0\%$).

Conclusions: The study shows that OAS plays a beneficial role in premenopausal women aged 40 years or younger and advanced stage breast cancer. However, OAS is associated with increase in hot flushes and vaginal dryness.

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Keywords: Breast cancer; Meta-analysis; Ovarian ablation; Ovarian suppression; Premenopausal

Introduction

Breast cancer is the most common cancer and is a leading cause of cancer-related death in women.¹ The

American Cancer Society estimates that about 231,840 new cases are diagnosed with breast cancer and 40,290 die of the disease among United States women in 2015.² Approximately 4% of women with breast cancer are younger than 40 years of age and 16% of age 40–49.¹ 60% of premenopausal women with breast cancer expresses estrogen receptors and/or progesterone receptors.³ Current adjuvant treatment of premenopausal women with breast cancer include selective estrogen receptor modulators (tamoxifen or toremifene), chemotherapy, ovarian

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ablation by radiotherapy or surgical oophorectomy, and ovarian suppression by luteinizing hormone-releasing hormone (LHRH) agonists (goserelin, triptorelin, leuprolide).⁴ Although ovarian ablation or suppression (OAS) has been widely used in premenopausal women with breast cancer, its effect is controversial in previous studies.^{5–7} Some studies showed positive results in favor of OAS, while others indicated inconsistent conclusions, and the American Society of Clinical Oncology endorsed guidelines recommending that OAS not be added routinely to adjuvant therapy in premenopausal women early stage invasive breast cancer.⁸

The Prevention of Early Menopause Study (POEMS) compared 105 premenopausal women treated with OAS plus chemotherapy with 113 experienced chemotherapy alone and demonstrated superior disease-free survival in those with OAS plus chemotherapy (HR 0.49, $P = 0.04$).⁷ A previous meta-analysis indicated that OAS significantly increased disease-free survival by 4.3% and overall survival by 3.2% at 15-years.⁹ Nevertheless, some other studies reported inconsistent findings. No statistically significant difference in disease-free survival for women treated with OAS plus chemotherapy and those who received chemotherapy alone was found in a randomized phase-3 PROMISE-GIM6 (Prevention of Menopause Induced by Chemotherapy: a Study in Early Breast Cancer Patients-Gruppo Italiano Mammella 6) study (HR 1.17, $P = 0.52$).¹⁰ The findings of Suppression of Ovarian Function Trial (SOFT) involving 3066 premenopausal women indicated that the addition of OAS to tamoxifen did not significantly improve disease-free survival (HR 0.83, $P = 0.10$) and overall survival (HR 0.74, $P = 0.13$), as compared with tamoxifen alone.¹¹ Thus we compared the effectiveness of OAS versus no OAS in premenopausal women with breast cancer to provide the latest evidence by reviewing the available studies. Meta-analytical methods were used to evaluate overall survival, disease-free survival and adverse events.

Materials and methods

Study selection

A literature review of EMBASE, Web of Science, PUBMED, and Cochrane Library between 1946 and Dec 2015 were conducted, and the relevant references in the list of all included studies were also identified. No language restriction was applied. The search terms included “breast cancer”, “premenopausal”, “ovarian function suppression”, “ovarian ablation”, “goserelin”, “triptorelin” and “leuprolide”. The MeSH search headings included: “breast neoplasms”, “premenopause”, “goserelin”, “oophorectomy”. Two authors independently reviewed the studies according to the included criteria. Study selection process was done and reported in according with the PRISMA guidelines.¹²

Study inclusion and exclusion criteria

The inclusion criteria included: (1) Design: randomized controlled trials (RCTs). (2) Patients: premenopausal women with breast cancer. (3) Intervention: OAS versus Tamoxifen, OAS plus Tamoxifen versus Tamoxifen, OAS plus Tamoxifen versus Chemotherapy, OAS plus Chemotherapy versus Chemotherapy, and OAS versus Chemotherapy. (4) Outcomes: disease-free survival, overall survival, and adverse events. We excluded the articles if they were: case reports, non-comparative studies, conference abstracts, reviews and studies with less than 30 patients.

Data extraction and management

The data were extracted by two authors independently in accordance with Cochrane guidelines. The extracted information included: first author, year of publication, trial acronym, recruitment period, eligibility, treatment arm, follow-up and outcomes. Disagreements were resolved through discussion.

Assessment of risk of bias

Two authors (Chang-Zai Li and Pan Zhang) assessed the risk of bias of the included studies according to the criteria outlined in the Cochrane Handbook for Systematic Reviews of Interventions.¹³ The assessed quality items would include (1) randomization sequence generation (checking for possible selection bias), (2) allocation concealment (checking for possible selection bias), (3) blinding (participants, investigators, outcome assessors and data analysis), (4) incomplete outcome data (checking for possible attrition bias through withdrawals, and dropouts), (5) selective outcome reporting and (6) other potential sources of bias.

Quality assessment

The Grading of Recommendations Assessment Development and Evaluation (GRADE) system was used to assess the quality of randomized controlled trials.¹⁴ We constructed a ‘Summary of finding’ table to evaluate the quality of evidence for the main review outcomes by using GRADEpro software. The evidence quality was classified into 4 levels: high ($\oplus\oplus\oplus\oplus$), moderate ($\oplus\oplus\oplus\ominus$), low ($\oplus\oplus\ominus\ominus$) or very low ($\oplus\ominus\ominus\ominus$), by considering the evidence directness, methodologic quality, effect estimates precision, data heterogeneity, and publication bias. The quality of the included studies was assessed independently.

Statistical analysis

This study was carried out under the guidance of the Cochrane Collaboration.¹⁵ For dichotomous outcome (adverse event), risk ratio (RR) was used as the summary statistic.

An RR of less than 1 favored the OAS group. For time-to-event outcomes (disease-free survival and overall survival), hazard ratio (HR) was assessed as the summary statistic. An HR less than 1 represented a survival benefit favoring the OAS group. A *P* value less than 0.05 was considered as statistically significant. The heterogeneity was tested by chi-square-based *Q* statistic. The heterogeneity was considered to be statistically significant at the *P* value less than 0.1. I-squared statistic (*I*²) measured the extent of inconsistency and represents the proportion of total variation. *I*² of less than 25% represents low heterogeneity, 25–50% as moderate heterogeneity, and more than 50% as high heterogeneity.¹⁶ A fixed-effects model was used if the heterogeneity was not significantly different. If not, the random-effects model was reported after exploring the cause of heterogeneity. The publication bias was tested by funnel plot. The results robustness to heterogeneity was assessed by sensitivity analyses. Funnel plot analyses were used to estimate publication bias. Analyses were done using Review Manager Software 5.3.

Subgroup analyses

OAS plus Tamoxifen versus Tamoxifen, OAS plus Tamoxifen versus Chemotherapy, OAS plus Chemotherapy versus Chemotherapy, OAS versus Chemotherapy, age (40 years or younger and older than 40 years), number of

positive lymph nodes (less or more than 4 positive lymph nodes), and tumor stage (early or advanced stage).

Results

Search results

We identified 754 studies by searching the databases. 283 duplicate studies were removed, and 471 potentially appropriate studies were identified and screened. 425 studies were excluded because they were case reports, non-comparative studies, conference abstracts, reviews and editorials. 46 potentially eligible studies remained for more detailed full-text evaluation. Among these, 15 studies had overlapping population, and the study with the most complete data was selected. 2 studies were excluded because the data were unavailable for analysis.^{17,18} Finally, this meta-analysis included 29 randomized controlled trials.^{7,10,11,19–44} Fig. 1 shows the selection process.

Study characteristics

Table 1 shows the main characteristic of the included trials. 29 studies with 21,249 patients were included from 1985 to 2015. Analysis was done on 9516 patients who had OAS and 9558 patients who underwent no OAS. The mean age was 43.2 years in OAS group and 43.4 years in

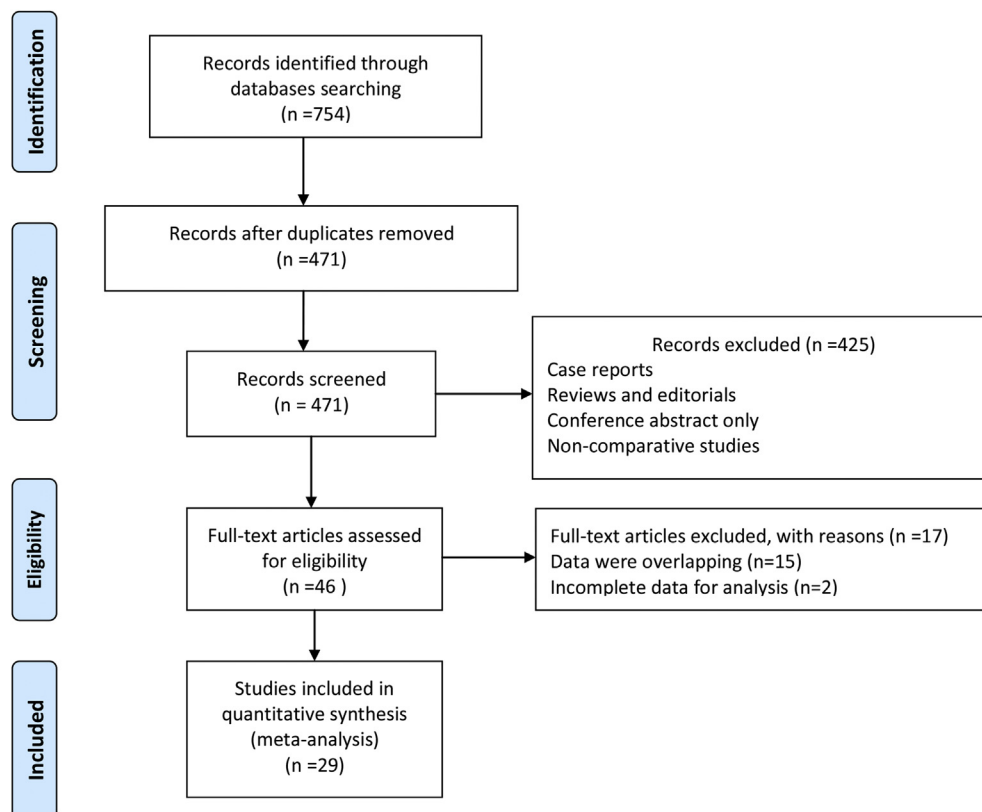


Figure 1. Flow chart of study selection process.

Table 1
Characteristic of included studies.

Study	Trial acronym	Patients		Recruitment period	Eligibility	Treatment arms	Follow-up (years)	Outcomes
		OAS	No OAS					
Lambertini 2015	PROMISE-GIM6	148	133	2003–2008	Pre, HR ⁻ /HR ⁺ , stage I–III	Triptorelin + Chemotherapy vs Chemotherapy	7.3	DFS
Moore 2015	POMS	105	113	2004–2011	Pre, HR ⁻ , stage I–III A	Goserelin + Chemotherapy vs Chemotherapy	4.1	OS, DFS, adverse event
Francis 2015	SOFT	1015	1018	2003–2011	Pre, HR ⁺ , early stage	OAS (triptorelin, surgical or radiation ablation) + Exemestane vs OAS (triptorelin, surgical or radiation ablation) + TAM 5 y vs TAM 5 y	5.6	OS, DFS, adverse event
Tevaarwer 2014	ECOG E3193	170	167	1994–1997	Pre, HR ⁺ , N ⁻ , T ≤ 3 cm	OAS (goserelin 5 y, leuprolide 5 y, surgical or radiation ablation) + TAM 5 y vs TAM 5 y	9.9	OS, DFS, adverse event
Schmid 2007	TABLE	294	295	1995–1998	Pre, HR ⁺ , N ⁺ , stage I–III A	Leuporelin 3-monthly 2 y vs CMF × 6 cycles	5.8	OS, DFS,
Kaufmann 2007	GABG trial IV- B-93	384	392	1993–2000	Pre, HR ⁻ /HR ⁺ , pT1–3, N0–N3, M0	Goserelin every 4 weeks 2 y vs CMF × 3 cycles for N0–N3/EC × 4 cycles + CMF × 3 cycles for N4–N9	4.7	OS, DFS, adverse event
Adjuvant 2007	ABC (OAS)	1063	1081	1993–2000	Pre, ER ⁻ /ER ⁺ , pT1–3a, N0–1, M0	OAS (goserelin 2 y, leuprolide 2 y, surgical or radiation ablation) + TAM vs TAM	5.9	OS, DFS
Von 2006	GABG trial IV-A-93	393	378	1993–2001	Pre, HR ⁺ , N ⁻	Goserelin every 4 weeks 2 y vs CMF × 3 cycles	4.9	OS, DFS
Roche 2006	FASG 06	164	169	1990–1998	Pre, HR ⁺ , N1–N3, early stage	Triptorelin 3 y + TAM 3 y vs FEC × 3cycles	7.0	OS, DFS
Ejlertsen 2006	DBCG 89B	376	386	1990–1998	Pre, HR ⁺ , N ⁺ , early stage	Ovarian ablation vs CMF × 4 cycles every 3 weeks	8.5	OS, DFS
Baum 2006	ZIPP	1354	1356	1987–1999	Pre, ER ⁻ /ER ⁺ , early stage	Goserelin every 4 weeks 2 y + TAM 2 y vs TAM 2 y or Goserelin every 4 weeks 2 y vs TAM	5.5	OS, DFS
Mitsuyama, 2005	ZBCSG Trial B	93	94	1994–1998	Pre, ER ⁺ , stage I, II, III A	Goserelin every 4 weeks 2 y + TAM 2 y vs TAM 2 y vs Goserelin every 4 weeks 2 y	4.0	OS, DFS
De 2005	GOCSI	120	114	1991–1996	Pre, ER ⁻ /ER ⁺ , pT0–3, N1–2, M0	CMF × 6 cycles vs doxorubicin × 4 cycles followed by CMF × 6 cycles vs CMF × 6 cycles followed by goserelin + TAM 2 y vs doxorubicin × 4 cycles followed by CMF × 6 cycles followed by goserelin + TAM 2 y	6.0	OS, DFS
Davidson 2005	INT 0101 (E5188)	502	494	1989–1994	Pre, HR ⁺ , N ⁺ ,	CAF × 6 cycles vs CAF followed by monthly goserelin 5 y vs CAF followed by monthly goserelin and daily TAM 5 y	9.6	OS, DFS, adverse event

Arriagada 2005	NA	461	465	1989–1997	Pre, ER ⁻ /ER ⁺ , N ⁺ , early stage	OAS (triptorelin 3 y, surgical or radiation ablation) + CT (CAF/CMF/FEC) vs CT (CAF/CMF/FEC)	9.5	OS, DFS
International 2003	IBCSG VIII	357	360	1990–1999	Pre, ER ⁻ /ER ⁺ , pT1–3, N0, M0	Goserelin 24 m vs CMF × 6 cycles vs CMF × 6 cycles + goserelin 18 m	7.0	OS, DFS
Söreide 2002	NA	159	161	1989–1994	Pre, N ⁺ , stage II	Goserelin 2 y vs TAM 2 y	7.3	OS
Jonat 2002	ZEBRA	797	817	1990–1996	Pre, ER ⁻ /ER ⁺ , N ⁺ , stage II	Goserelin 2 y vs CMF × 6 cycles	6.0	OS, DFS
Jakesz 2002	ABCSG 5	511	523	1990–1999	Pre, ER ⁻ /ER ⁺ , stage I and II	Goserelin 3 y + TAM 5 y vs CMF × 6 cycles	5.0	OS, DFS
Klijn 2000	EORTC	53	54	1988–1995	Pre, advanced breast cancer	Buserelin + TAM vs buserelin vs TAM	7.3	OS, DFS
Boccardo 2000	IBCASG 02	124	120	1989–1997	Pre, ER ⁺ , early stage	OAS (oophorectomy, ovarian irradiation, and goserelin) + TAM 5 y vs CMF × 6 cycles	6.3	OS, DFS
Sawka 1997	NCIC CTG	19	20	1982–1984	Pre, advanced breast cancer	Ovarian ablation vs TAM	NA	OS
Rivkin 1996	ASOGS	148	140	1979–1989	Pre, ER ⁺ , N ⁺	Oophorectomy + CMFVP vs CMFVP	7.7	OS, DFS
Bartlett 1993	Scottish	167	165	1980–1990	Pre, ER ⁻ /ER ⁺ , pT0–3, N0–2, M0	Ovarian ablation vs CMF every 3 weeks	5.9	OS, DFS
Goldhirsch 1990	IBCSG	177	179	1978–1991	Pre, ER ⁻ /ER ⁺ , N ⁺	Oophorectomy + CMF vs CMF	8.0	OS, DFS
Falkson 1987	ECOG	100	100	1968–1983	Pre, ER ⁺ , advanced breast cancer	Oophorectomy + CAF vs CAF	NA	OS
Ingle 1986	NA	27	26	1978–1984	Pre, advanced breast cancer	Oophorectomy vs TAM	NA	OS, DFS
Buchanan 1986	NA	58	59	1979–1983	Pre, advanced breast cancer	Oophorectomy vs TAM	NA	OS
Aron 1985	LBCS	177	179	1978–1981	Pre, pT1–3, N0–1, M0	Oophorectomy + CMF vs CMF	4.0	OS, DFS

Abbreviations: OAS: ovarian ablation or suppression CT: chemotherapy OS: overall survival DFS: disease-free survival ER⁺: estrogen receptor positive ER⁻: estrogen receptor negative HR⁺: hormone receptor positive. HR⁻: hormone receptor negative Pre: premenopausal TAM: Tamoxifen N⁺: node positive N⁻: node negative y: years m: month T: tumor vs: versus CMF: cyclophosphamide, methotrexate and fluorouracil CAF: cyclophosphamide, doxorubicin, and fluorouracil FEC: 5-fluorouracil, doxorubicin or epidoxorubicin, and fluorouracil CNF: cyclophosphamide mitoxantrone (Novantrone), and fluorouracil NA: not available CMFVP: cyclophosphamide, methotrexate, fluorouracil, vincristine, and prednisone.

no OAS group. The lymph node status was not significantly different between two groups ($P = 0.74$). There were significant differences in tumor size ($P < 0.01$) and grading ($P = 0.04$) between OAS and no OAS (Table 2).

Risk of bias in included studies

All studies were described as randomized, but the risk of bias of 16 RCTs^{7,20–26,28,30,33,34,36–38,41} was unclear, and they did not provide an adequate description of the methods used for random sequence generation. 26 RCTs^{7,10,11,19–26,28–34,36–38,40–44} did not provide an adequate description of the methods used for generating the allocation sequence, we considered that these included studies carried an unclear risk of bias. Two RCTs^{37,42} that did not do double blind had high risk of bias in performance bias and detection bias. 11 RCTs^{19–22,24–26,28,31,33,36} provided no relevant description, nor information regarding dropouts, so we considered these studies to be at unclear risk of bias. We found no protocols for 21 of the studies,^{11,19–28,31–38,41,42} and were therefore unable to assess reporting bias. We did not recognize any other potential sources of bias for 9 studies^{19–22,24–26,31,36} (Fig. 2).

Effects of interventions on premenopausal breast cancer

OAS versus tamoxifen

6 studies investigating 3268 patients reported overall survival^{20,21,26,31,36,37} (Fig. 3a). A significant higher overall survival was found in OAS than Tamoxifen (HR 0.99, 95% CI: 0.95–1.03, $P = 0.64$). There was no significant heterogeneity ($\chi^2 = 2.64$, $P = 0.76$, $I^2 = 0\%$).

Data on disease-free survival was available in 3 studies evaluating 2950 patients^{21,36,37} (Fig. 4a). The difference between the two groups was not significant (HR 1.04, 95% CI: 0.96–1.12, $P = 0.33$), and the study heterogeneity was also not significant ($\chi^2 = 0.35$, $P = 0.84$, $I^2 = 0\%$).

Table 2

Baseline characteristics between OAS and no OAS in premenopausal breast cancer.

Characteristic	OAS	no OAS	P
Mean age (years)	43.2 (37.6–46.0)	43.4 (38.7–46.0)	0.21
Tumor size			<0.01
T1	3334 (49.7%)	3399 (49.3%)	
T2	3212 (46.5)	3093 (45.0%)	
T3–4	253 (3.3%)	398 (5.7%)	
Lymph node status			0.74
N0	925 (17.7)	959 (18.2%)	
N1–3	2930 (55.9%)	2951 (55.7%)	
N4+	1386 (26.4%)	1372 (26.1%)	
Grading			0.04
1	499 (15.0%)	406 (12.3%)	
2	1735 (52.2%)	1804 (54.5%)	
3	1092 (32.8%)	1098 (33.2%)	

Abbreviation: OAS: ovarian ablation or suppression.

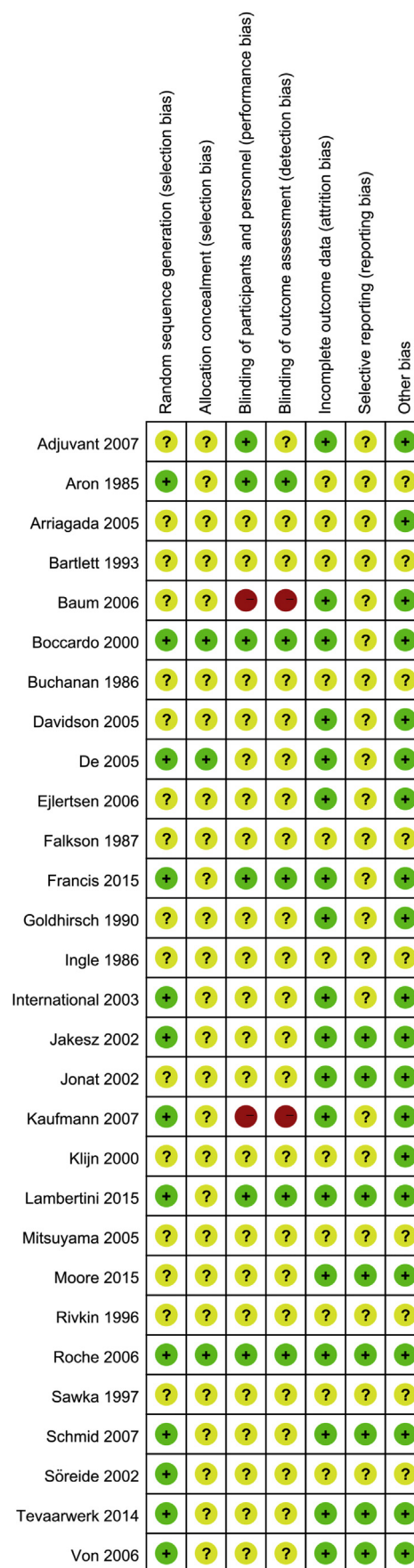
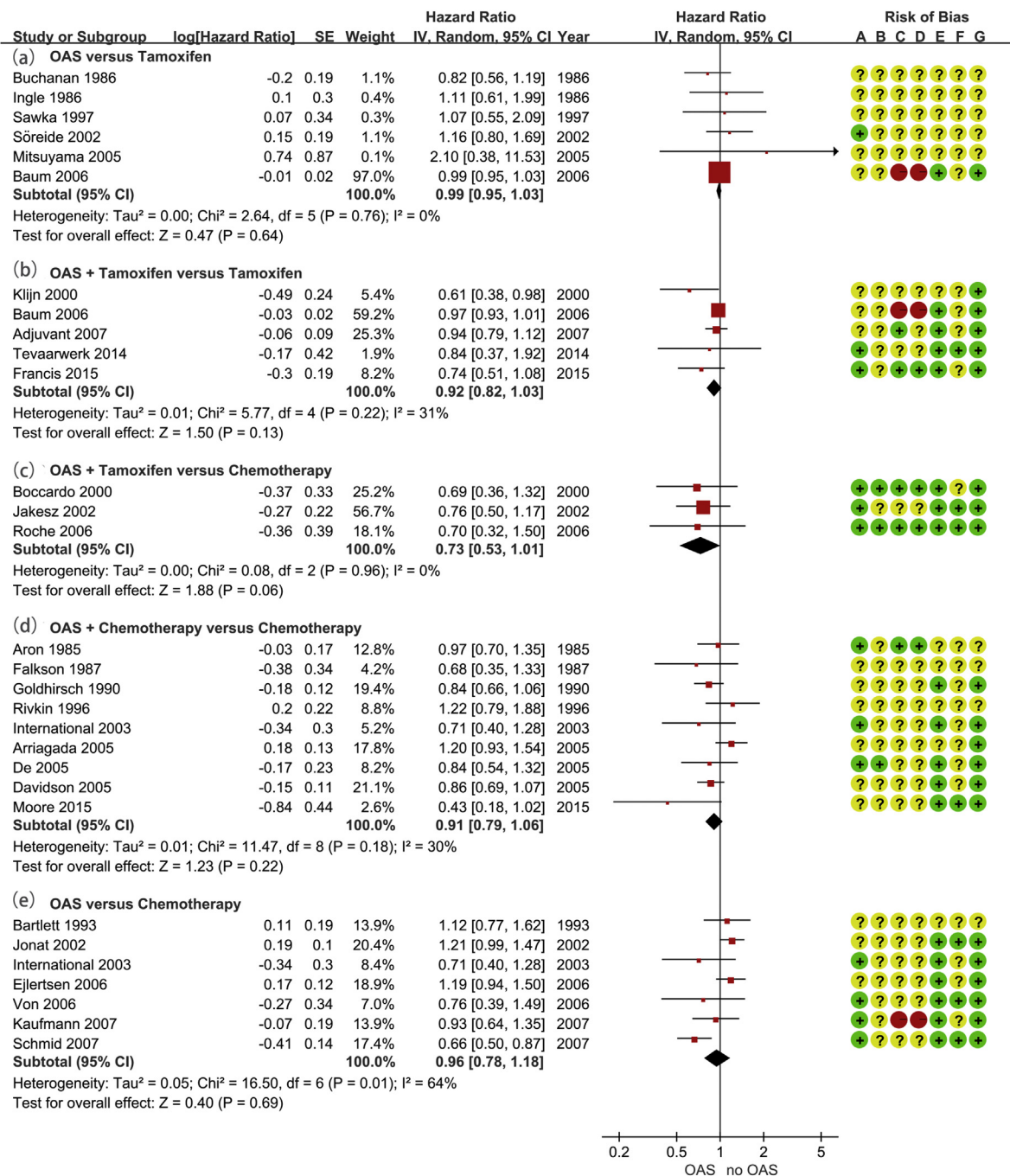


Figure 2. Risk of bias summary.

**Risk of bias legend**

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

- Low risk of bias
 Unclear risk of bias
 High risk of bias

Figure 3. Forest plot of overall survival. a: OAS versus Tamoxifen. b: OAS plus Tamoxifen versus Tamoxifen. c: OAS plus Tamoxifen versus Chemotherapy. d: OAS plus Chemotherapy versus Chemotherapy. e: OAS versus Chemotherapy. Abbreviations OAS: ovarian ablation suppression; CI: confidence interval; SE: standard error; IV: Inverse Variance.

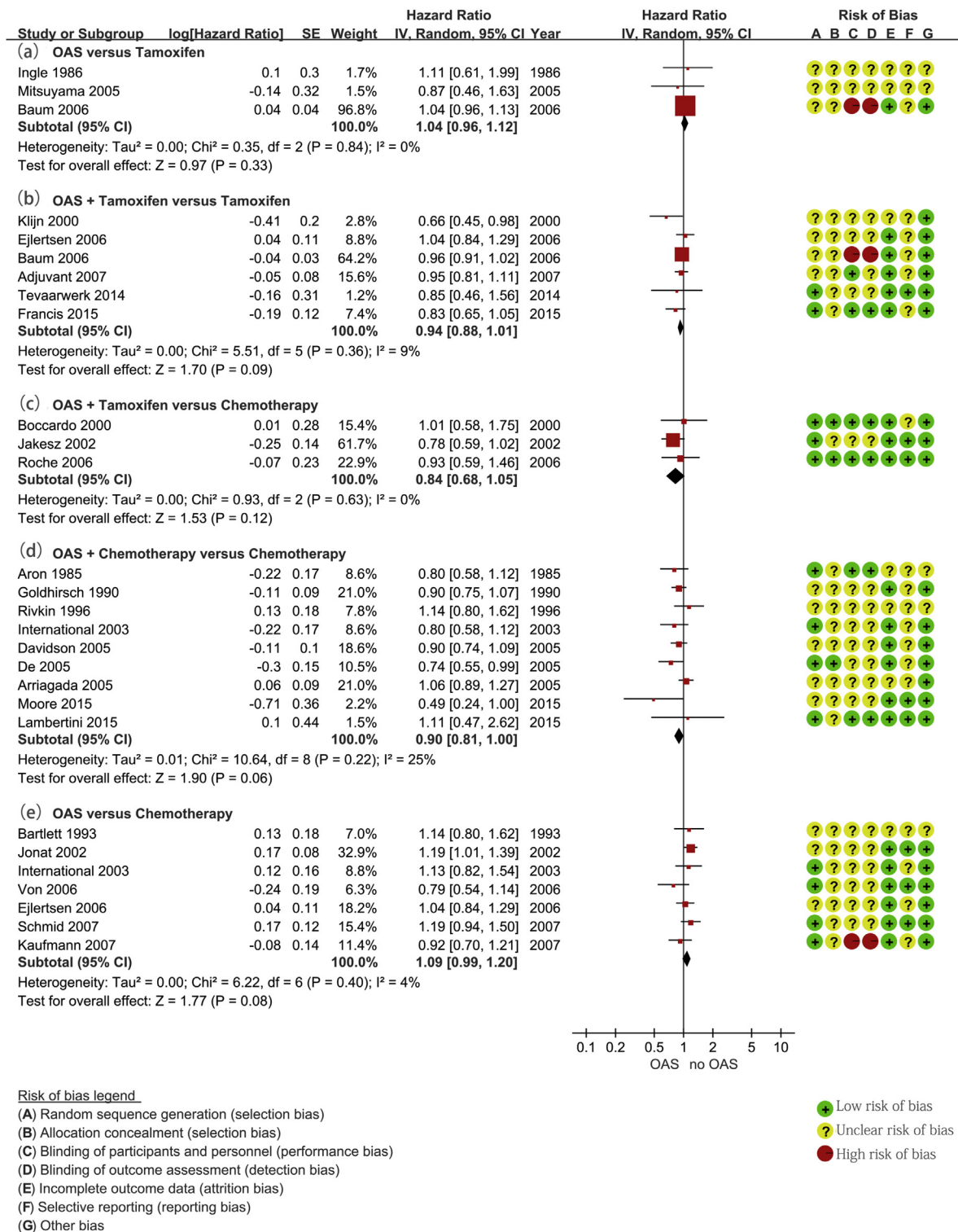


Figure 4. Forest plot of disease-free survival. a: OAS versus Tamoxifen. b: OAS plus Tamoxifen versus Tamoxifen. c: OAS plus Tamoxifen versus Chemotherapy. d: OAS plus Chemotherapy versus Chemotherapy. e: OAS versus Chemotherapy. Abbreviations: OAS: ovarian ablation suppression; CI: confidence interval; SE: standard error; IV: Inverse Variance.

OAS plus tamoxifen versus tamoxifen

5 studies assessed overall survival in 7331 patients,^{11,28,37,41,44} and the difference between the two

groups was not significant (HR 0.92, 95% CI: 0.82–1.03, $P = 0.13$; Fig. 3b), with moderate between-study heterogeneity ($\chi^2 = 5.77$, $P = 0.22$, $I^2 = 31\%$). We therefore performed a sensitivity analysis in which we removed the

study at a high risk of bias.³⁷ The overall survival was not substantially altered (HR 0.84, 95% CI: 0.69–1.01, $P=0.07$). Between-study heterogeneity was markedly reduced by the sensitivity analysis ($\chi^2 = 3.63$, $P=0.30$, $I^2 = 17\%$).

A total of 6 studies reported data for disease-free survival.^{11,28,38,41,44} The difference between the two groups was not significant (HR 0.94, 95% CI: 0.88–1.01, $P=0.09$; Fig. 4b), with no evidence of statistically significant heterogeneity between studies ($\chi^2 = 5.51$, $P=0.36$, $I^2 = 9\%$).

OAS plus tamoxifen versus chemotherapy

Overall survival data was available for 1611 patients across 3 studies.^{27,29,39} The Chemotherapy group reported higher HR than the OAS plus Tamoxifen group, but no statistically significant difference was found (HR 0.73, 95% CI: 0.53–1.01, $P=0.06$; Fig. 3c), and the between-study heterogeneity was not significant ($\chi^2 = 0.08$, $P=0.96$, $I^2 = 0\%$).

Data on disease-free survival was extracted from 3 studies^{27,29,39} assessing 1611 patients (Fig. 4c). There was no significant difference between the two groups in disease-free survival (HR 0.84, 95% CI: 0.68–1.05, $P=0.12$), with no significant between-study heterogeneity ($\chi^2 = 0.93$, $P=0.63$, $I^2 = 0\%$).

OAS plus chemotherapy versus chemotherapy

Data on overall survival was available in 9 studies evaluating 4291 patients^{7,19,22,23,25,32–35} (Fig. 3d). The difference between the two groups was not significant (HR 0.91, 95% CI: 0.79–1.06, $P=0.22$), with moderate between-study heterogeneity ($\chi^2 = 11.47$, $P=0.18$, $I^2 = 30\%$). In a sensitivity analysis, removing the studies with high risks of bias^{22,25} did not markedly decrease the pooled estimate (HR 0.90, 95% CI: 0.77–1.05, $P=0.18$), where heterogeneity remained moderate ($\chi^2 = 9.01$, $P=0.17$, $I^2 = 33\%$).

9 studies^{7,10,19,23,25,32–35} that assessed 4372 patients reported disease-free survival (Fig. 4d). No significant difference for disease-free survival was found between the two groups (HR 0.90, 95% CI: 0.81–1.00, $P=0.06$), with low between-study heterogeneity ($\chi^2 = 10.64$, $P=0.22$, $I^2 = 25\%$).

OAS versus chemotherapy

Data on overall survival was extracted from 7 studies assessing 5561 patients^{24,30,32,38,40,42,43} (Fig. 3e). No significant difference between two groups was found (HR 0.96, 95% CI: 0.78–1.18, $P=0.69$), and the between-study heterogeneity was significant ($\chi^2 = 16.50$, $P=0.01$, $I^2 = 64\%$). A further sensitivity analysis was performed by excluding the study with a high risk of bias.⁴² The

overall survival was not substantially altered (HR 0.96, 95% CI: 0.75–1.22, $P=0.73$), where heterogeneity remained statistically significant ($\chi^2 = 16.27$, $P=0.006$, $I^2 = 69\%$).

Disease-free survival was reported in 7 studies^{24,30,32,38,40,42,43} assessing 5561 patients (Fig. 4e). The difference between the two groups was not significant (HR 1.09, 95% CI: 0.99–1.20, $P=0.08$), and the heterogeneity between studies was not significant ($\chi^2 = 6.22$, $P=0.40$, $I^2 = 4\%$).

Effects of age on premenopausal breast cancer

A subgroup analyses found that premenopausal women aged 40 years or younger had higher disease-free survival (HR 0.84, 95% CI: 0.73–0.97, $P=0.02$; $I^2 = 0\%$) and overall survival (HR 0.78, 95% CI: 0.66–0.94, $P=0.008$; $I^2 = 0\%$) when they underwent OAS. There was evidence of publication bias, with asymmetry in the funnel plot as shown in Supplementary Fig. 1. In women older than 40 years, no significant differences were found for disease-free survival (HR 0.91, 95% CI: 0.80–1.03, $P=0.15$; $I^2 = 30\%$) and overall survival (HR 0.88, 95% CI: 0.75–1.02, $P=0.10$; $I^2 = 0\%$) between the two groups. No publication bias was detected by funnel plot (Supplementary Fig. 1).

Effects of axillary lymph nodes on premenopausal breast cancer

No significant differences for disease-free survival (HR 1.08, 95% CI: 0.91–1.29, $P=0.35$; $I^2 = 0\%$) and overall survival (HR 0.96, 95% CI: 0.75–1.23, $P=0.74$; $I^2 = 22\%$) were found among women with four or more positive lymph nodes between two groups were found. Data on overall survival were extracted from 10 studies^{19,32,34,35,38–41,43,44} assessing one to three positive lymph nodes. The difference between the two groups was not significant (HR 0.92, 95% CI: 0.81–1.04, $P=0.17$), and the between-study heterogeneity was not significant ($\chi^2 = 9.72$, $P=0.37$, $I^2 = 7\%$). 11 studies^{19,27,32,34,35,38–41,43,44} assessed disease-free survival for patients with one to three positive lymph nodes. The results favored the no OAS group (HR 0.90, 95% CI: 0.82–0.98, $P=0.02$), with no evidence of statistically significant heterogeneity between studies ($\chi^2 = 6.11$, $P=0.81$, $I^2 = 0\%$) (Supplementary Table 1).

Effects of tumor stage assignment on premenopausal breast cancer

16 studies^{19,21–24,27,30–34,37–41,44} assessed the overall survival for premenopausal women with early stage breast cancer. The difference between the two groups was not significant (HR 0.93, 95% CI: 0.86–1.00, $P=0.06$), with low between-study heterogeneity ($\chi^2 = 17.63$, $P=0.28$,

$I^2 = 15\%$). 16 studies^{11,19,24,27,29,30,32,33,35,37–39,41,42,44} reported disease-free survival, and the results favored the OAS group (HR 0.90, 95% CI: 0.84–0.96, $P = 0.002$) with no significant heterogeneity between studies ($\chi^2 = 16.99$, $P = 0.32$, $I^2 = 12\%$). No publication bias was detected by funnel plot (Supplementary Fig. 2).

Data on overall survival was extracted from 5 studies^{20–22,26,28} assessing advanced stage breast cancer, and showed a significant difference favoring the OAS group (HR 0.76, 95% CI: 0.60–0.96, $P = 0.02$), with no significant between-study heterogeneity ($\chi^2 = 2.09$, $P = 0.72$, $I^2 = 0\%$). 2 studies^{21,28} assessed disease-free survival for patients with advanced stage breast cancer. The difference between the two groups was not significant (HR 0.82, 95% CI: 0.50–1.33, $P = 0.41$), with evidence of between-study heterogeneity ($\chi^2 = 2.00$, $P = 0.16$, $I^2 = 50\%$). There was evidence of publication bias, with asymmetry in the funnel plot as shown in Supplementary Fig. 2.

Adverse event

6 studies^{7,11,20,29,43,44} that assessed 4307 patients reported hot flushes (Supplementary Fig. 3a). No OAS group found statistically significantly less hot flushes (risk ratio, 1.91, 95% CI: 1.62–2.26, $P < 0.01$), and the between-study heterogeneity was not significant ($\chi^2 = 4.10$, $P = 0.54$, $I^2 = 0\%$).

4 studies^{7,11,20,43} that assessed 3373 patients reported depression (Supplementary Fig. 3b). The difference between the two groups was not significant (RR 1.28, 95% CI: 0.94–1.74, $P = 0.12$), with no significant heterogeneity between studies ($\chi^2 = 2.57$, $P = 0.46$, $I^2 = 0\%$).

2 studies^{7,11} that assessed 2225 patients reported vaginal dryness (Supplementary Fig. 3c). No OAS group found statistically significantly less vaginal dryness (RR 1.19, 95% CI: 1.08–1.31, $P = 0.0003$). Between-study heterogeneity was no significant ($\chi^2 = 0.20$, $P = 0.65$, $I^2 = 0\%$).

Heterogeneity analysis

There was significant heterogeneity in overall survival ($\chi^2 = 16.50$, $P = 0.01$, $I^2 = 64\%$). The heterogeneity may be related to clinical diversity, such as the differences of treatment schedule. The Takeda Adjuvant Breast Cancer Study with Leuprorelin acetate (TABLE) trial⁴³ was designed to compare the 3-monthly LHRH agonist leuprorelin acetate (LAD-3M) and chemotherapy with cyclophosphamide, methotrexate, and fluorouracil (CMF) in premenopausal breast cancer, and results showed that overall survival analysis favored LAD-3M (HR 1.50, 95% CI: 1.13–1.99, $P = 0.005$). However, a number of other trials^{30,40,42} have directly evaluated the efficacy of ovarian suppression with monthly goserelin versus chemotherapy with CMF in premenopausal breast cancer, and have showed that no significant difference was found in overall

survival between goserelin and CMF. After we removed the TABLE trial,⁴³ overall survival was not substantially altered (HR 1.10, 95% CI: 0.97–1.26, $P = 0.14$). Between-study heterogeneity was significantly reduced after the heterogeneity analysis ($\chi^2 = 5.28$, $P = 0.38$, $I^2 = 5\%$).

Discussion

This study compared the effects of OAS and no OAS on overall survival, disease-free survival, and adverse effects in premenopausal women with breast cancer. On subgroup analyses of OAS versus chemotherapy, OAS plus chemotherapy versus chemotherapy, OAS versus tamoxifen, OAS plus tamoxifen versus tamoxifen, and OAS plus tamoxifen versus chemotherapy, no significant differences were found in disease-free survival and overall survival. On subgroup analyses of age, a benefit for OAS in disease-free survival and overall survival in premenopausal women aged 40 years or younger was found, but no significant differences for women older than 40 years were showed. On subgroup analyses of stage, there was a statistically significant benefit of OAS versus no OAS in terms of overall survival for advanced stage breast cancer. On subgroup analyses of axillary lymph nodes, no significant differences in terms of disease-free survival and overall survival were seen. OAS was associated with more hot flushes and vaginal dryness.

Previous meta-analysis compared the differences between OAS and no OAS in premenopausal breast cancer, but the effect of OAS was controversial. A Cochrane analysis published in 2009 found that LHRH agonists could be an effective adjuvant therapy for women with early stage breast cancer. In 2005, the Early Breast Cancer Trialists' Collaborative Group (EBCTCG) systematic overview of randomized trials assessed the effects of chemotherapy and hormonal therapy for early breast cancer on recurrence and 15-year survival. The result showed that, compared with chemotherapy alone, OAS in addition to chemotherapy significantly improved disease-free survival and overall survival in premenopausal women aged less than 50 years.^{9,45} However, this overview leaves several important questions unanswered. First, it is unclear as to whether the combination of OAS and tamoxifen is more effective than tamoxifen alone. Second, whether the effect of OAS in the risk reduction for women younger than 40 years of age was different from those of age 40–49 is unclear. Our review answered the above questions in some extent. First, we compared the OS and DFS between OAS plus Tamoxifen and Tamoxifen alone, and the results showed that there were no significant differences between two groups in OS and DFS. The findings of a recently published meta-analysis were consistent with our study. This meta-analysis included 4 RCTs totaling 6279 premenopausal breast cancer, and the pooled data showed that no significant differences in terms of DFS (RR 0.92; 95% CI 0.84–1.00; $P = 0.06$) and OS (RR 0.91; 95% CI

0.80–1.03; $P = 0.15$) between OAS plus tamoxifen and tamoxifen alone.⁶ Second, we performed a subgroup analysis by age on premenopausal women. Our review indicated that premenopausal women aged 40 years or younger had higher DFS and OS in OAS group. In women older than 40 years, no significant differences were found for DFS and OS between the two groups. Our results were consistent with a meta-analysis⁵ by Cuzick et al., which concluded that, comparing with no systemic treatment, LHRH agonists did not significantly reduce recurrence ($P = 0.08$) or death after recurrence ($P = 0.49$). The addition of LHRH agonists to tamoxifen, chemotherapy, or both significantly reduced recurrence, death after recurrence, and all deaths in women aged 40 years or younger, but no significant difference for premenopausal women older than 40 years were found. Besides, there were several significant differences in our review from previously published meta-analysis. First, our meta-analysis, including 29 RCTs with 21,249 women, increased the sample size, and thus, the results accuracy. Second, our review found that OAS was associated with a substantial increase in terms of hot flushes and vaginal dryness, and no significant difference was found in depression, which was not assessed in previous meta-analysis. Third, evidence quality was assessed by the GRADE system, and most of the evidence of quality was moderate (Supplementary Table 2). In addition to, we performed subgroup analyses according to interventions, age, number of positive lymph nodes, and tumor stage.

The review has at least four limitations. First, publication bias is detected with asymmetrical funnel plot advanced stage subgroup (Supplementary Fig. 2). The high quality prospective trials on larger populations are needed to further confirm this result. Second, the heterogeneity of the overall survival has been shown to be significant in OAS versus Chemotherapy subgroup ($\chi^2 = 16.50$, $P = 0.01$, $I^2 = 64\%$). The differences in sample size, patient age, tumor stage, grade, treatment arms and other factors among the studies might be related to the heterogeneity. The heterogeneity analysis indicated that clinical diversity result in heterogeneity, such as the differences of treatment schedule. Third, evidence quality was low in OAS plus Chemotherapy versus Chemotherapy and OAS versus Chemotherapy subgroups (Supplementary Table 2). Although several included studies were described as randomized control trials they did not describe the random sequence method, and there were inconsistency among studies. Therefore, we downgraded the quality of evidence to low. We should explain the results prudently. Fourth, the inclusion of only English studies may lead to potential publication bias.

Conclusion

This study shows that the OAS plays a beneficial role in premenopausal women aged 40 years or younger and advanced stage breast cancer. Compared with no OAS,

OAS is associated with a substantial increase in hot flushes and vaginal dryness. Due to the potential publication bias and study heterogeneity in several aspects, the results should be interpreted with caution. Further high quality randomized controlled trials are needed.

Conflict of interest statement

The authors have no conflicts of interest to declare.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.ejso.2016.11.011>.

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